

AWS ACM + TLS/SSL Termination on Application Load Balancer (ALB)

Complete Detailed Guide

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1. Introduction to SSL/TLS

What is SSL?

SSL stands for Secure Sockets Layer.

It was the older technology used for encrypting communication between:

- Browser ↔ Server
- Client ↔ API
- User ↔ Website

Example:

Without SSL:

User → Internet → Website

Data can be read by attackers.

With SSL/TLS:

User → Encrypted Tunnel → Website

Data becomes encrypted.

What is TLS?

TLS stands for Transport Layer Security.

TLS is the modern and secure replacement for SSL.

Today people commonly say:

- SSL Certificate
- SSL/TLS

But technically:

- Modern websites use TLS
 - SSL is outdated
-

2. What is ACM?

AWS Certificate Manager (ACM)

AWS ACM is a service used to:

- Create SSL/TLS certificates
- Manage certificates
- Renew certificates automatically
- Attach certificates to AWS services

Supported services:

- Application Load Balancer (ALB)
 - CloudFront
 - API Gateway
 - Network Load Balancer (TLS listener)
-

Why Use ACM?

Without ACM:

You must:

- Buy certificate
- Generate CSR manually
- Upload private key
- Renew manually
- Handle expiration

With ACM:

AWS automatically:

- Issues certificate
 - Validates ownership
 - Renews certificate
 - Manages private keys
-

3. What is SSL/TLS Termination?

Definition

TLS termination means:

The Load Balancer handles HTTPS encryption and decryption.

Example:

Browser → HTTPS → ALB → HTTP → EC2

The ALB decrypts HTTPS traffic.

This process is called:

- SSL Offloading
 - TLS Termination
-

Why Use TLS Termination?

Benefits:

1. EC2 instances don't handle encryption
 2. Reduced CPU usage on servers
 3. Easier certificate management
 4. Centralized HTTPS handling
 5. Easy renewal using ACM
 6. Better scalability
-

4. Why Use TLS Termination on ALB?

Real Scenario

Suppose you have:

- 10 EC2 instances
- One website
- HTTPS enabled

Without ALB:

You must install certificate on all 10 servers.

With ALB:

Only ALB needs certificate.

Flow:

User → HTTPS → ALB → HTTP → EC2

Very easy to manage.

5. Architecture Flow

Full Flow

User Browser ↓ HTTPS 443 Application Load Balancer ↓ HTTP 80 Target Group ↓ EC2 Instances

HTTPS Flow

1. User visits:

<https://example.com>

1. Browser contacts ALB
 2. ALB presents ACM certificate
 3. TLS handshake happens
 4. Encrypted connection established
 5. ALB decrypts traffic
 6. ALB forwards request to EC2
-

6. Prerequisites

You need:

- AWS Account
 - Domain Name
 - Running EC2 Instance
 - Web server installed
 - Security groups configured
 - Public subnet for ALB
-

7. Domain Purchase and DNS Basics

What is a Domain?

Example:

- google.com
- amazon.com
- mywebsite.com

You can purchase domains from:

- Route 53
 - GoDaddy
 - Namecheap
 - Cloudflare
-

DNS Basics

DNS converts:

Domain → IP Address

Example:

example.com → 54.22.11.90

Important DNS Records

A Record

Maps:

Domain → IP Address

Example:

example.com → 54.11.22.33

CNAME Record

Maps:

Subdomain → Another Domain

Example:

www.example.com → abc.elb.amazonaws.com

ACM validation also uses CNAME records.

8. Step 1 — Request SSL Certificate in ACM

Open ACM Console

Go to:

AWS Console → Certificate Manager (ACM)

Click Request Certificate

Choose:

Request a public certificate

Click:

Next

Add Domain Names

Example:

example.com

Optional wildcard:

*.example.com

Wildcard covers:

- app.example.com
- api.example.com
- dev.example.com

Validation Method

Choose:

DNS Validation

Why DNS validation?

- Easier
- Automatic renewal supported
- Recommended by AWS

Key Algorithm

Usually keep default:

RSA 2048

Click:

Request

9. Step 2 — Domain Validation using CNAME

After request:

Certificate status:

Pending validation

ACM Provides CNAME Details

Example:

Name:

_abcd123.example.com

Value:

_xyz987.acm-validations.aws

Type:

CNAME

10. Add CNAME Record in DNS

If Using Route 53

Click:

Create records in Route 53

AWS automatically creates:

CNAME validation record.

Manual DNS Example

If using GoDaddy or Namecheap:

Add record:

Type: CNAME

Host: _abcd123

Points to: _xyz987.acm-validations.aws

TTL: 300

Save.

11. Understanding How CNAME Validation Works

What Actually Happens?

AWS wants proof that:

You own the domain.

So ACM checks:

"Does this DNS record exist?"

If record exists:

AWS confirms ownership.

Then:

Certificate becomes ISSUED.

Internal Flow

1. ACM generates unique token
 2. You add token as CNAME
 3. ACM checks DNS globally
 4. Ownership verified
 5. Certificate issued
-

12. Verify Certificate Issued Status

Go to ACM console.

Status should become:

Issued

This may take:

- Few minutes
- Sometimes 30 minutes

Important

Certificate and ALB must be in same AWS region.

Example:

- ALB in us-east-1
- ACM certificate must also be in us-east-1

Otherwise certificate won't appear.

13. Step 5 — Create Application Load Balancer

Go to:

EC2 → Load Balancers

Click:

Create Load Balancer

Choose:

Application Load Balancer

Basic Configuration

Name

Example:

prod-alb

Scheme

Internet-facing

IP Address Type

IPv4

Network Mapping

Choose:

- VPC
- At least 2 public subnets

Example:

- Public subnet A
 - Public subnet B
-

14. Security Group for ALB

Create security group:

Allow inbound:

HTTP 80 → 0.0.0.0/0 HTTPS 443 → 0.0.0.0/0

Outbound:

Allow all

Attach this SG to ALB.

15. Step 6 — Configure HTTPS Listener

Listener Basics

Listener checks:

- Port
 - Protocol
-

Add Listeners

HTTP Listener

Protocol: HTTP

Port: 80

HTTPS Listener

Protocol: HTTPS

Port: 443

16. Step 7 — Attach ACM Certificate to ALB

While creating HTTPS listener:

Choose:

Default SSL/TLS certificate

Select:

Certificate from ACM

Choose your domain certificate.

Example:

example.com

Security Policy

Choose:

ELBSecurityPolicy-TLS13-1-2-2021-06

This defines:

- TLS versions
 - Cipher suites
-

Important

When attached:

ALB now handles:

- HTTPS
 - Encryption
 - Decryption
 - TLS handshake
-

17. Step 8 — Configure Target Group

Create Target Group

Target type:

Instances

Protocol

HTTP

Port

80

Health Check Path

/

Register Targets

Add EC2 instances.

18. Step 9 — Register EC2 Instances

Select:

- Web server instances

Click:

Include as pending below

Create target group.

19. Step 10 — Security Group Configuration

ALB Security Group

Inbound:

80 from anywhere 443 from anywhere

Outbound:

All traffic

EC2 Security Group

Inbound:

HTTP 80 from ALB security group

Important:

Source should be:

ALB Security Group

NOT:

0.0.0.0/0

Why?

This ensures:

Only ALB can access EC2.

Direct internet access blocked.

20. Step 11 — Test HTTPS Access

Copy ALB DNS.

Example:

myalb-123456.us-east-1.elb.amazonaws.com

Open browser:

<https://myalb-123456.us-east-1.elb.amazonaws.com>

If Using Custom Domain

Create Route 53 Alias Record.

Example:

example.com → ALB

Route 53 Alias Record Steps

Go to:

Route 53 → Hosted Zone

Create Record

Type: A Record

Enable: Alias

Choose:

Application Load Balancer

Select your ALB.

Save.

21. Step 12 — Redirect HTTP to HTTPS

Why Redirect?

If user types:

http://example.com

We should automatically redirect to:

https://example.com

Configure Redirect

Edit HTTP listener.

Add action:

Redirect to HTTPS

Settings:

Protocol: HTTPS Port: 443 Status code: HTTP_301

22. SSL Handshake Explained

Step-by-Step TLS Handshake

Step 1

Browser says:

"I want secure connection"

Step 2

ALB sends certificate.

Certificate contains:

- Public key
 - Domain name
 - Expiry date
 - CA details
-

Step 3

Browser validates:

- Is certificate trusted?
 - Is domain matching?
 - Is certificate expired?
-

Step 4

Browser and ALB generate session keys.

Step 5

Encrypted communication starts.

23. What Happens Internally During TLS Termination

Detailed Flow

1. Browser sends encrypted request
 2. ALB decrypts request
 3. ALB reads HTTP headers
 4. ALB applies rules:
 5. Path-based routing
 6. Host-based routing
 7. WAF checks
 8. ALB forwards request to target group
 9. EC2 sends response
 10. ALB encrypts response
 11. Browser receives encrypted data
-

24. Difference Between End-to-End Encryption and TLS Termination

TLS Termination

Browser → HTTPS → ALB → HTTP → EC2

Encryption ends at ALB.

End-to-End Encryption

Browser → HTTPS → ALB → HTTPS → EC2

Encryption continues till EC2.

When Use End-to-End?

Use when:

- Compliance requirements
 - Sensitive banking data
 - Internal encryption required
-

25. ACM Automatic Certificate Renewal

Huge Advantage of ACM

AWS automatically renews certificate.

No manual renewal needed.

Renewal Requirements

DNS validation CNAME must remain present.

Do NOT delete:

ACM validation CNAME.

Otherwise renewal fails.

26. Common Errors and Troubleshooting

Error 1

Certificate not visible in ALB.

Reason:

Different region.

Fix:

Create ACM certificate in same region.

Error 2

Certificate pending validation.

Reason:

CNAME incorrect.

Fix:

- Check host
 - Check value
 - Wait DNS propagation
-

Error 3

HTTPS not working.

Reason:

Security group blocking 443.

Fix:

Allow inbound HTTPS.

Error 4

502 Bad Gateway

Reason:

Backend server not responding.

Fix:

- Check EC2
 - Check Apache/Nginx
 - Check health checks
-

Error 5

Target unhealthy.

Reason:

Health check failing.

Fix:

- Verify app running
 - Check path
 - Check SG rules
-

27. Real-Time Request Flow Example

Scenario

User visits:

<https://shop.example.com/products>

Complete Flow

1. DNS resolves domain
2. Browser gets ALB IP
3. Browser connects to ALB on 443
4. TLS handshake occurs
5. ACM certificate validated
6. Secure session established
7. ALB decrypts request
8. ALB checks listener rules
9. Request forwarded to EC2
10. EC2 sends response
11. ALB encrypts response

12. Browser receives HTTPS response

28. Best Practices

Use DNS Validation

Preferred for automatic renewal.

Use HTTPS Only

Redirect HTTP to HTTPS.

Restrict EC2 Access

Allow traffic only from ALB SG.

Use Latest TLS Policies

Avoid old TLS versions.

Enable Access Logs

Useful for debugging and auditing.

Use WAF with ALB

Protect from:

- SQL injection
 - XSS
 - Bots
-

29. Interview Questions

Q1. What is ACM?

AWS Certificate Manager is used to provision, manage, and renew SSL/TLS certificates.

Q2. What is TLS termination?

ALB decrypts HTTPS traffic and forwards plain HTTP traffic to backend servers.

Q3. Why use ACM?

- Automatic renewal
 - Easy integration
 - No manual certificate management
-

Q4. Why use DNS validation?

Supports automatic renewal and easier validation.

Q5. Can ACM certificate be exported?

Public ACM certificates cannot be exported.

Q6. Why should EC2 SG allow ALB SG only?

For better security and private backend access.

30. Summary

What We Learned

- SSL vs TLS
- ACM basics
- DNS validation
- CNAME verification

- ALB HTTPS listener
 - TLS termination
 - HTTP to HTTPS redirect
 - Security group setup
 - TLS handshake
 - Automatic certificate renewal
 - Best practices
-

31. Complete Practical Lab (End-to-End Setup)

Goal

We will create:

- EC2 web server
- Application Load Balancer
- ACM certificate
- HTTPS website
- HTTP to HTTPS redirect
- Route 53 domain mapping

Final Result:

<https://example.com>

working securely using ACM.

Lab Architecture

Internet User ↓ HTTPS 443 ↓ ALB ↓ HTTP 80 ↓ EC2 Apache Server

Step 1 — Launch EC2 Instance

Go to:

EC2 → Launch Instance

Configuration

Name

web-server-1

AMI

Ubuntu Server 24.04 LTS

Instance Type

t2.micro

Key Pair

Select existing key pair.

Network Settings

Choose:

- VPC
- Public subnet

Enable:

Auto assign public IP

Security Group for EC2

Inbound Rules:

SSH 22 → Your IP HTTP 80 → ALB Security Group

Important:

Do NOT allow:

HTTP 80 → 0.0.0.0/0

because only ALB should access EC2.

User Data Script

Paste this in User Data.

```
#!/bin/bash
apt update -y
apt install apache2 -y
systemctl start apache2
systemctl enable apache2

cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Secure Website</title>
</head>
<body>
<h1>Website Working Through ALB HTTPS</h1>
<h2>Hostname: $(hostname)</h2>
</body>
</html>
EOF
```

Launch instance.

Step 2 — Verify Web Server

Open:

http://EC2-Public-IP

You should see:

Website Working Through ALB HTTPS

Step 3 — Create ACM Certificate

Go to:

AWS Console → Certificate Manager

Request Certificate

Click:

Request Certificate

Choose:

Request a public certificate

Domain Name Example

example.com

Optional:

*.example.com

Validation Method

Select:

DNS validation

Click:

Request

Step 4 — Understand the Validation Screen

After requesting certificate:

Status:

Pending Validation

You will see:

Domain Name	Validation Status	CNAME Name	CNAME Value
-------------	-------------------	------------	-------------

Example:

Name:

_abcd123.example.com

Value:

_xyz987.acm-validations.aws

Why ACM Gives CNAME?

AWS must verify:

"You actually own this domain."

So AWS asks:

"Add this secret DNS record."

If you can add it:

You own the domain.

Step 5 — Add CNAME Record in Route 53

Go to:

Route 53 → Hosted Zones

Select your domain.

Create Record

Type:

CNAME

Record Name:

_abcd123

Value:

_xyz987.acm-validations.aws

TTL:

300

Save.

Important DNS Note

Sometimes users mistakenly add:

_abcd123.example.com.example.com

This is wrong.

If hosted zone already is:

example.com

then only add:

_abcd123

Step 6 — Wait for Validation

ACM continuously checks DNS.

Once CNAME is found:

Status changes to:

Issued

What Happens Internally?

1. ACM queries public DNS
 2. Searches for CNAME
 3. Validates ownership
 4. Generates certificate
 5. Stores certificate securely
 6. Certificate ready for ALB
-

Step 7 — Create Target Group

Go to:

EC2 → Target Groups

Click:

Create Target Group

Configuration

Target Type:

Instances

Name

web-target-group

Protocol

HTTP

Port

80

Health Check Path

/

Register Targets

Select EC2 instance.

Click:

Include as pending below

Create target group.

Step 8 — Understand Health Checks

ALB periodically sends requests:

GET /

If server responds:

200 OK

then target becomes:

Healthy

If Health Check Fails

Possible reasons:

- Apache not running
- Wrong port
- Security group blocked

- Wrong health path
 - App crashed
-

Step 9 — Create ALB

Go to:

EC2 → Load Balancers

Click:

Create Load Balancer

Choose:

Application Load Balancer

Basic Configuration

Name

secure-web-alb

Scheme

Internet-facing

IP Type

IPv4

Network Mapping

Choose:

- VPC
- Public subnet AZ1
- Public subnet AZ2

Why 2 subnets?

ALB requires minimum 2 Availability Zones.

Step 10 — Create ALB Security Group

Inbound:

HTTP 80 → 0.0.0.0/0 HTTPS 443 → 0.0.0.0/0

Outbound:

All traffic

Attach SG to ALB.

Step 11 — Configure Listener

HTTP Listener

Protocol: HTTP

Port: 80

Action: Redirect to HTTPS

Redirect Settings

Protocol: HTTPS Port: 443 Status Code: HTTP_301

Why HTTP_301?

301 means:

Permanent Redirect.

Browsers remember this redirect.

Step 12 — Configure HTTPS Listener

Protocol: HTTPS

Port: 443

Default Action

Forward to:

web-target-group

Step 13 — Attach ACM Certificate

In HTTPS Listener:

Choose:

From ACM

Select:

example.com certificate

SSL Policy

Choose:

ELBSecurityPolicy-TLS13-1-2-2021-06

What SSL Policy Controls

Defines:

- TLS versions
- Encryption ciphers
- Security strength

Step 14 — Create ALB

Click:

Create Load Balancer

Wait until:

State becomes Active.

Step 15 — Test ALB DNS

Copy ALB DNS.

Example:

secure-web-alb-123.us-east-1.elb.amazonaws.com

Test HTTP

Open:

http://ALB-DNS

Expected:

Automatically redirected to HTTPS.

Test HTTPS

Open:

https://ALB-DNS

You should see:

Secure padlock icon.

Why Browser Shows Secure Lock?

Because:

- Certificate trusted
 - Domain valid
 - Encryption enabled
-

Step 16 — Configure Route 53 Domain Mapping

Go to:

Route 53 → Hosted Zone

Create Record.

Record Type

A Record

Enable:

Alias

Alias Target

Select:

Application Load Balancer

Choose ALB.

Save.

Final Test

Open:

https://example.com

Now traffic flow becomes:

User → Route53 → ALB → EC2

Step 17 — Verify TLS Certificate in Browser

Open website.

Click lock icon.

You can see:

- Certificate issuer
 - Expiry date
 - Domain name
 - Encryption details
-

Step 18 — Understand TLS Termination Deeply

Browser to ALB

Encrypted HTTPS.

ALB to EC2

Usually plain HTTP.

Why Decrypt at ALB?

ALB needs to:

- Read headers
- Read paths
- Apply routing rules
- Apply WAF
- Perform authentication

Encrypted traffic cannot be inspected.

Example

User requests:

`https://example.com/api/users`

ALB decrypts request.

Then checks:

IF path starts with `/api` → send to API servers

Otherwise: → send to frontend servers

Step 19 — Multiple Certificates on Same ALB

ALB supports SNI.

Meaning:

One ALB can host:

- `app.example.com`
- `api.example.com`
- `admin.example.com`

Each with different certificate.

Step 20 — ACM Renewal Process

Before expiry:

ACM automatically:

1. Generates new certificate
2. Validates DNS CNAME
3. Renews certificate
4. Updates ALB automatically

No downtime.

Important Production Best Practices

Use Private EC2 Instances

Production setup usually:

Internet → ALB → Private EC2

EC2 should not have public IP.

Use Auto Scaling

Attach ASG to target group.

Enable Access Logs

Useful for:

- Auditing
 - Monitoring
 - Troubleshooting
-

Use AWS WAF

Protects from:

- SQL Injection
 - XSS
 - Bots
 - Rate attacks
-

Use HTTPS Only

Never expose production apps over plain HTTP.

Important Real Interview Scenario

Question

Why terminate SSL at ALB instead of EC2?

Answer

Because:

- Centralized certificate management
 - Reduced CPU load on servers
 - Easier scaling
 - Automatic ACM renewal
 - Easier operations
 - ALB can inspect requests for routing
-

Complete Flow Summary

1. User enters domain
 2. Route 53 resolves domain
 3. Browser connects to ALB
 4. TLS handshake occurs
 5. ALB presents ACM certificate
 6. Secure HTTPS established
 7. ALB decrypts traffic
 8. ALB checks routing rules
 9. Request sent to target group
 10. EC2 processes request
 11. Response sent back to ALB
 12. ALB encrypts response
 13. Browser receives HTTPS response
-

Final Architecture

User Browser ↓ HTTPS 443 Application Load Balancer ↓ HTTP 80 Target Group ↓ EC2 Instances

32. Deep Understanding of Certificates

What is Inside a Certificate?

An SSL/TLS certificate contains:

- Domain name
 - Public key
 - Certificate authority information
 - Expiry date
 - Digital signature
 - Encryption details
-

Public Key vs Private Key

Public Key

Shared with everyone.

Used for:

Encryption.

Stored inside certificate.

Private Key

Secret.

Used for:

Decryption.

AWS ACM securely stores private key.

You never directly access ACM private keys.

Certificate Authority (CA)

A CA is a trusted organization.

Examples:

- Amazon Trust Services
 - DigiCert
 - Let's Encrypt
 - Sectigo
-

Why Browser Trusts ACM Certificate

Browsers already trust:

Amazon Trust Services.

So when ALB presents ACM certificate:

Browser accepts it.

Self-Signed Certificate vs ACM Certificate

Self-Signed

You create certificate yourself.

Browser warning appears.

Example:

"Your connection is not private"

ACM Certificate

Trusted publicly.

No warnings.

Wildcard Certificates

Example

*.example.com

Covers:

- api.example.com
- app.example.com
- dev.example.com

Does NOT cover:

example.com

You must explicitly add:

example.com

SAN Certificates

SAN = Subject Alternative Name.

One certificate can cover:

- example.com
 - api.example.com
 - admin.example.com
-

RSA vs ECDSA Certificates

RSA

Most common.

Highly compatible.

ECDSA

Modern.

Faster.

Smaller keys.

Better performance.

But some older clients may not support it.

33. Deep Dive into HTTPS Listener

What is a Listener?

Listener waits for incoming traffic.

Example:

HTTPS Listener:

- Port 443
 - Protocol HTTPS
-

Listener Components

A listener contains:

- Protocol
 - Port
 - SSL certificate
 - Rules
 - Default actions
-

Listener Rules

Rules decide:

Where traffic should go.

Example Path-Based Routing

`/api/` → API servers `/images/` → Image servers `/*` → Frontend servers

Example Host-Based Routing

`api.example.com` → API target group `app.example.com` → Frontend target group `admin.example.com` → Admin target group

Rule Priority

ALB checks rules from top to bottom.

First matching rule wins.

Default Rule

If no rules match:

Traffic goes to default target group.

34. Deep Dive into Target Groups

What is a Target Group?

Target group is a collection of backend servers.

Examples:

- EC2 instances
 - IP addresses
 - Lambda functions
-

Types of Targets

Instance Targets

Registers EC2 instance IDs.

IP Targets

Registers private IP addresses.

Useful for:

- On-premises servers
 - Containers
-

Lambda Targets

ALB can invoke Lambda functions.

Health Check Detailed Flow

ALB continuously monitors targets.

Example:

Every 30 seconds:

GET /

Healthy Threshold

Suppose:

Healthy threshold = 5

Then:

5 successful health checks required.

Unhealthy Threshold

Suppose:

Unhealthy threshold = 2

Then:

2 failed checks mark target unhealthy.

Why Health Checks Matter

If one server crashes:

ALB automatically stops sending traffic.

35. Security Best Practices

Use HTTPS Everywhere

Never expose login pages over HTTP.

Restrict Backend Access

EC2 should accept traffic only from ALB SG.

Use Private Subnets for EC2

Production architecture:

Internet → Public ALB → Private EC2

Use Bastion Host or SSM

For secure SSH access.

Enable Access Logs

Store ALB logs in S3.

Useful for:

- Security auditing
 - Traffic analysis
 - Troubleshooting
-

Enable CloudWatch Metrics

Monitor:

- Request count
 - Latency
 - Errors
 - Healthy hosts
-

Use AWS Shield

Protects against:

DDoS attacks.

36. TLS Versions Explained

TLS 1.0

Old.

Not secure.

Deprecated.

TLS 1.1

Also outdated.

TLS 1.2

Widely used today.

Secure.

TLS 1.3

Latest version.

Benefits:

- Faster handshake
 - Better security
 - Improved performance
-

Why Disable Old TLS Versions?

Older versions contain vulnerabilities.

Attackers may exploit them.

37. Understanding Encryption

Symmetric Encryption

Same key used for:

- Encryption

- Decryption

Fast.

Used after TLS handshake.

Asymmetric Encryption

Uses:

- Public key
- Private key

Used during TLS handshake.

Why TLS Uses Both?

Asymmetric:

Secure key exchange.

Symmetric:

Fast communication.

38. ALB vs NLB TLS Termination

ALB

Layer 7.

Understands:

- HTTP
- HTTPS
- Headers
- Cookies
- Paths

Supports:

- Path routing
 - Host routing
 - WAF
-

NLB

Layer 4.

Works with:

- TCP
- UDP
- TLS

Very high performance.

Can preserve source IP.

TLS Termination in NLB

NLB can also terminate TLS.

But:

NLB cannot inspect HTTP paths.

Real Scenario

Use ALB when:

- Web applications
- Path routing
- Host routing

Use NLB when:

- Ultra high performance
- Static IP required
- Source IP preservation needed

39. ACM with CloudFront

Why Use ACM with CloudFront?

For global HTTPS delivery.

Important Rule

CloudFront certificates must be in:

us-east-1

Flow

User → CloudFront HTTPS → ALB/Origin

Benefits

- CDN acceleration
 - Lower latency
 - Edge caching
 - HTTPS globally
-

40. ACM with API Gateway

API Gateway supports:

Custom HTTPS domains.

Example:

api.example.com

using ACM certificate.

41. End-to-End HTTPS Architecture

Architecture

Browser → HTTPS → ALB → HTTPS → EC2

Why Use This?

Required for:

- Compliance
 - Banking systems
 - Healthcare
 - Highly secure systems
-

How to Configure

1. Install HTTPS server on EC2
 2. Use port 443 target group
 3. ALB forwards HTTPS traffic
 4. Backend also uses certificates
-

42. Common ACM and HTTPS Interview Questions

Q1. Difference between SSL and TLS?

TLS is modern secure version of SSL.

Q2. What is TLS termination?

ALB decrypts HTTPS traffic before forwarding.

Q3. Why use ACM?

Automatic certificate management and renewal.

Q4. Why DNS validation preferred?

Supports automatic renewal.

Q5. What happens if validation CNAME deleted?

Certificate renewal fails.

Q6. Why use ALB instead of configuring certificates on EC2?

Centralized management and easier scaling.

Q7. Difference between ALB and NLB?

ALB works at Layer 7. NLB works at Layer 4.

Q8. Why HTTPS is important?

Protects data confidentiality and integrity.

Q9. Why browser shows certificate warning?

Certificate untrusted or mismatched.

Q10. What is wildcard certificate?

One certificate securing multiple subdomains.

43. Real Production Architecture Example

Architecture

Users ↓ Route 53 ↓ CloudFront ↓ AWS WAF ↓ ALB ↓ Auto Scaling Group ↓ Private EC2 Instances ↓ RDS Database

HTTPS Flow in Production

1. User accesses domain
 2. Route53 resolves domain
 3. CloudFront edge receives request
 4. HTTPS handshake occurs
 5. WAF filters attacks
 6. ALB routes traffic
 7. EC2 processes request
 8. Database responds
 9. Response encrypted back to user
-

44. Final Important Notes

Important Note 1

ACM public certificates are free.

Important Note 2

Public ACM certificates cannot be exported.

Important Note 3

Certificate region must match ALB region.

Important Note 4

Never delete DNS validation CNAME.

Important Note 5

Always redirect HTTP to HTTPS.

Important Note 6

Use latest TLS policies.

Important Note 7

Use private EC2 in production.

Important Note 8

ALB itself does not store website files.

It only routes traffic.

Important Note 9

Target group decides where traffic goes.

Important Note 10

Security groups are critical for backend protection.

Complete End-to-End Flow Recap

1. User enters domain
 2. DNS resolves ALB
 3. Browser connects on 443
 4. TLS handshake occurs
 5. ACM certificate validated
 6. Encrypted session established
 7. ALB decrypts request
 8. Listener rules checked
 9. Request forwarded to target group
 10. EC2 processes request
 11. Response returned to ALB
 12. ALB encrypts response
 13. Browser receives secure response
-

Key Takeaway

ACM + ALB provides:

- Secure HTTPS communication
- Easy certificate management
- Automatic renewal
- Scalable architecture
- Simplified operations
- Centralized TLS handling